Spring Data Access – Assessment Questions

1. Consider a Spring application that has a Employee bean with employee id, name and employee qualifications. Write an Employee DAO interface that has a method to insert employee information to the DB (Assumption: EMPLOYEE table in DB with columns – Employee ID, Employee Name and Employee Qualifications)
   Implement the EmployeeDAO and utilize datasource alone to perform the insert operation. Externalize the database properties in a configuration file (eg: spring-datasources.xml) in addition to the basic bean definitions configuration. Utilize both the xml files in your spring application

2. Re-modify the above scenario to use JDBCTemplate and the update method to insert the employee information to the DB. The datasource and JDBCTemplate are the attributes of the DAO implementation class

3. Modify your spring application of inserting employee information to DB using JdbcDaoSupport.
   a. Change the DAO implementation to extend the JdbcDaoSupport
   b. Remove the attributes of datasource and JdbcTemplate. Utilize the getJdbcTemplate() to process the SQL query

4. Enhance the spring application to process the below operations
   a. Query a single employee based on employee id
   b. Query for a list of employees whose name starts with 'Ra'.
   c. Query for a list of employees whose qualitifcations match 'BE'

5. Modify EMPLOYEE table in DB to include a new column (Employee Blood Group). Include update functionality to the spring application to update the blood group for a given employee. The employee id should be the key for the search criteria.

6. Delete an employee from the DB based on employee id.